

BANARAS HINDU UNIVERSITY

B.Sc. Semester II Sessional Examination 2024-25 (NEP)

SEC (DST-CIMS)**Paper Code: CIMSE21 : Computational Mathematics using SageMath
(Sessional)**

Time: 1 Hour

Maximum Marks: 20

Instructions: Question 1 is compulsory and is of 15 marks. All other questions carry equal marks. Solve any one question from Question 2 to 3.

Question 1. Write the codes in SageMath for the following mathematical operations and then solve the corresponding question without using codes: [15]

- (a) Inverse of a non-singular matrix and find the inverse of the matrix $A = \begin{pmatrix} 5 & 3 \\ 3 & 2 \end{pmatrix}$.
- (b) Orthogonal projection of a vector u on a vector w and find the orthogonal projection of vector $(2, 3, 4)$ on $(1, 1, 1)$.
- (c) Norm of a vector with respect to norms $\|\cdot\|_1$ and $\|\cdot\|_2$ and hence compute the two norms for the vector $v = (2, -3, 6)$.
- (d) Area of the parallelogram formed by two vectors and hence find area of parallelogram formed by vectors $(1, 1, 1)$ and $(1, 1, 0)$.
- (e) Echelon form of a matrix and hence find the echelon form of the matrix $A = \begin{pmatrix} 5 & 3 \\ 3 & 2 \end{pmatrix}$.

Question 2. Find the remainder when 2^{2025} is divided by 17. Write the codes in SageMath to find the remainder. [5]

Question 3. Write the code in SageMath to plot a function $f : I \rightarrow \mathbb{R}$, where I is an interval in $\mathbb{R}$. Can you plot a circle centered at the point $(1, 1)$ and radius 1 as a graph of a function? If yes, write the function with its SageMath code, if not, is there any way to plot this circle? Write its SageMath code and a possible diagram on your answer sheet. [5]
